METRIC

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MILITARY SPECIFICATION

ALUMINUM OXIDE, ANHYDROUS, ANALYZED REAGENT (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of aluminum oxide.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial).
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 101 - Preservation, Packaging, and Packing Materials: Test Procedures.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" by 48" Pallets.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Appearance. Aluminum oxide shall be a white powder free from foreign matter when tested as specified in 4.2.4.1.

3.2 Chemical characteristics. Aluminum oxide shall conform to the chemical characteristics of table I when tested as specified therein.

TABLE I. Chemical characteristics

Characteristic	Maximum percent by weight	Test paragraph
Chloride	0.010	4.2.4.2
Silicate	Trace	4.2.4.3
Sulfate	0.070	4.2.4.4
Alkalies and alkaline earths	0.50	4.2.4.5
Heavy metals (as Pb)	0.007	4.2.4.6
Iron	0.015	4.2.4.7
Loss on ignition (H_2O)	5.0	4.2.4.8
Water soluble substances	0.75	4.2.4.9

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the aluminum oxide produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.

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4.2.2.2 For aluminum oxide tests. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 25 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for aluminum oxide tests

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled and closed unit pack, intermediate pack, or shipping container, as applicable, ready for shipment. Sample unit packs, intermediate packs, and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Partitions or filler pads inadequate or not as specified
(f) Marking incorrect, missing or illegible
(g) Unitization not as specified

4.2.3.2 For aluminum oxide tests. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 2.5 percent defective.

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4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Appearance. Visually examine the specimen for color, form, and the presence of foreign matter.

4.2.4.2 Chloride.

(a) Chloride for control. Dissolve 0.0165 g of sodium chloride in water to make 1000 milliliters (ml). One ml contains 0.01 milligram (mg) of chloride.

(b) Procedure. Digest 1.000 g of the specimen with 45 ml of water and 5 ml of 70-percent nitric acid for 30 minutes and filter. To 25 ml of the filtrate add 1 ml of 2-percent silver nitrate solution. Compare any resulting turbidity with that in a control to which 5.0 ml of the solution prepared in (a) has been added. The turbidity of the specimen solution shall be no greater than that of the control.

4.2.4.3 Silicate. Fuse 0.50 g of the specimen with 10 g of potassium bisulfate for 1 hour in a platinum crucible, cool, and dissolve in 100 ml of hot water. No visible insoluble matter shall remain.

4.2.4.4 Sulfate.

(a) Sulfate for control. Dissolve 0.1814 g of potassium sulfate in water to make 1000 ml. One ml contains 0.1 mg of sulfate.

(b) Procedure. Boil 2.000 g of the specimen with 10 ml of 37-percent hydrochloric acid and 50 ml of water for 30 minutes, dilute with 130 ml of water and add an excess of 28-percent ammonium hydroxide. Boil until nearly free from ammonia odor, cool, dilute to 200 ml and filter. Neutralize 20 ml of the filtrate with 1N hydrochloric acid, then add 1 ml excess of the acid and 2 ml of 12-percent barium chloride solution. Compare the turbidity produced with that produced by a control made with 1.4 ml of the solution prepared in (a), 1 ml of 1N hydrochloric acid, and 2 ml of 12-percent barium chloride solution in the same final volume as the specimen. The turbidity of the specimen solution shall be no greater than that of the control.

4.2.4.5 Alkalies and alkaline earths. Evaporate 50 ml of the filtrate from the test for sulfate in 4.2.4.4(b) with 5 drops of 96-percent sulfuric acid and ignite. Cool, and weigh to the nearest 0.1 mg. Calculate the percent by weight alkalies and alkaline earths as follows:

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$$\text{Percent alkalies and alkaline earths} = \frac{400A}{W}$$

where: A = Weight of residue in grams, and
W = Weight of specimen in 4.2.4.4 in grams.

4.2.4.6 Heavy metals.

(a) Lead for control. Dissolve 0.0160 g of lead nitrate in water to make 1000 ml. One ml contains 0.01 mg of lead.

(b) Procedure. To 4.000 g of the specimen add 10 ml of 37-percent hydrochloric acid and 10 ml of water and evaporate to dryness on a steam bath. Heat the residue with 50 ml of water, filter, and wash with water to 80 ml. To 10 ml of the filtrate add 7.0 ml of the solution prepared in (a), dilute to 30 ml with water, and add 1 ml of 1N acetic acid (solution A). To another 30-ml aliquot of the filtrate, add 1 ml of 1N acetic acid (solution B). Then to each add 10 ml of a freshly prepared saturated aqueous solution of hydrogen sulfide. Solution B shall be no darker than solution A.

4.2.4.7 Iron.

(a) Butanolic potassium thiocyanate solution. Dissolve 10 g of potassium thiocyanate in 10 ml of water. Warm the solution to 25° to 30°C, dilute to 100 ml with n-butanol, and shake vigorously until clear.

(b) Iron for control. Dissolve 0.0864 g of ferric ammonium sulfate in 10 ml of 10-percent sulfuric acid and dilute with water to 1000 ml. One ml contains 0.01 g of iron.

(c) Procedure. Dilute 5.0 ml of the filtrate from the test for heavy metals in 4.2.4.6(b) with 5 ml of water, add 1 ml of 37-percent hydrochloric acid and 15 ml of the butanolic potassium thiocyanate solution prepared in (a), shake well for 30 seconds, then allow to separate. Compare any red color in the clear butanol layer with that of a control made with 3.8 ml of the solution prepared in (b), 1 ml of 37-percent hydrochloric acid, and 15 ml of the butanolic potassium thiocyanate solution prepared in (a) and diluted with water to the same final volume as the specimen. The color of the specimen solution shall be no darker than that of the control.

4.2.4.8 Loss on ignition. Weigh to the nearest milligram approximately 1 g of the specimen. Ignite strongly, cool, and weigh to the nearest milligram. Calculate the percent by weight loss on ignition as follows:

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$$\text{Percent loss on ignition} = \frac{100(W - A)}{W}$$

where: A = Weight of residue in grams, and
W = Weight of specimen in grams.

4.2.4.9 Water soluble substances. Weigh to the nearest milligram approximately 2 g of the specimen. Digest with 100 ml of water on a steam bath for 30 minutes, cool, dilute to 100 ml with water and filter. Evaporate 50 ml of the filtrate, ignite gently, cool, and weigh to the nearest 0.1 mg. Calculate the percent by weight water soluble substances as follows:

$$\text{Percent water soluble substances} = \frac{200A}{W}$$

where: A = Weight of residue in grams, and
W = Weight of specimen in grams.

4.2.5 Container leakage test. Conduct the container leakage test in accordance with Fed. Test Method Std. No. 101, method 5009.1, hot water technique.

5. PACKAGING

5.1 Unit packing. Aluminum oxide shall be unit packed level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. A quantity of 454 (+5 or -0) g of aluminum oxide shall be unit packed level A in a bottle furnished with a lined screw cap and pilfer-deterrent outer seal. The bottle shall conform to PPP-C-186, group A, class 1, type optional, style 2, grade optional with closure A, B, P or R, and with outer seal A. The closure shall be secured to the bottle at a torque within a range of 2.25 to 3.39 newton-meters. There shall be no evidence of leakage as indicated by the appearance of a steady stream of bubbles in the test bath water when tested as specified in 4.2.5 prior to applying the outer seal.

5.1.1.1 Intermediate packing. Twelve bottles of aluminum oxide unit packed as specified in 5.1.1 shall be intermediate packed level A upright in a close-fitting fiberboard box furnished with full bottle height partitions. The box shall conform to PPP-B-636, style RSC-L, grade W5c or W6c. The partitions shall form a close-fitting cell for each unit pack, and their flutes shall be parallel to the vertical dimension of the box. Motion of the contents of the box shall be prevented by the insertion, as needed, of fiberboard filler pads and partitions formed from the same material as that of the box. The box shall be closed as specified in PPP-B-636 for closure of interior containers.

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5.1.2 Level B. The aluminum oxide shall be unit packed as specified in 5.1.1 except that the bottle shall conform to PPP-C-186, group A, class 2, style 2, grade optional with closure A, B, P or R. It shall be furnished with outer seal A. This bottle shall be used as an alternative to the bottle specified in 5.1.1 at the option of the contractor.

5.1.2.1 Intermediate packing. Twelve bottles of aluminum oxide shall be intermediately packed level B as specified for level A protection in 5.1.1.1, except that the box shall conform to PPP-B-636, style RSC-L, class domestic, variety SW, grade 125.

5.1.3 Commercial. A quantity of aluminum oxide as specified (see 6.2) shall be unit packed in accordance with MIL-STD-1188.

5.2 Packing. Aluminum oxide shall be packed level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Two intermediate packs of aluminum oxide shall be packed level A upright in a close-fitting, nailed wood, cleated-plywood, or wirebound wood box for a type 2 load. The nailed wood box shall conform to PPP-B-621, class 2, style 4, grade A. The cleated plywood box shall conform to PPP-B-601, type overseas, style A, B or I, grade A. The wirebound wood box shall conform to PPP-B-585, class 3, style optional. Motion of contents within the box shall be prevented by the insertion as needed, of fiberboard filler pads formed from material conforming to PPP-F-320, grade V3c, W5c, or W6c. Each box shall be closed and reinforced as specified in the applicable box specification.

5.2.2 Level B. Two intermediate packs of aluminum oxide shall be packed level B in the manner specified for level A packing in 5.2.1, except that the box shall conform to PPP-B-636, style RSC-L, grade V3c, and reinforcement of the closed box shall not be required.

5.2.3 Commercial. Aluminum oxide shall be packed in accordance with MIL-STD-1188.

5.3 Unitization. Level A and B packs of aluminum oxide shall be palletized in accordance with MIL-STD-147, load type I using strapping, shrink film, or stretch film bonding. The pallet shall conform to NN-P-71, type IV and the pallet for level A packs shall conform to grade B treated, while that for level B packs shall conform to grade A untreated.

5.4 Marking. Level A and B unit packs, intermediate packs, packs, and palletized loads shall be marked in accordance with MIL-STD-129. Commercial unit packs and packs shall be marked in accordance with MIL-STD-1188. Each unit pack, intermediate pack, and pack shall be labeled or marked to show the following information:

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(a) Analysis, including maximum impurities
(b) Date of manufacture of the contents
(c) Lot or batch number of the contents

6. NOTES

6.1 Intended use. Aluminum oxide is intended for use as a chemical reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of unit packing and packing required (see 5.1 and 5.2)
(c) Unit quantity required (see 5.1.3)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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